2. Finite Difference Methods

Introduction

Though this textbook focuses on finite-volume methods, we first provide a brief review of finite-difference approximations. We focus our discussion on the forward, backward, and centered difference formulas for approximating first-order spatial derivatives, such as $d\phi/dx$, because these are used extensively in finite-volume methods. We also leverage the simple setting of finite differences to introduce key concepts related to stencils, error measurement, and order-of-accuracy. For completeness, we include an optional section discussing additional finite-difference formulas for estimating first- and second-order derivatives. Though less important to finite-volume methods, the formulas are useful for post-processing and verification. The derivation of the formulas using weighted combinations of Taylor series is also a general tool worth understanding.

2.1 Approximating the first derivative

Suppose a function $\phi(x)$ is known only at discrete points. This arises when $\phi(x)$ is measured, simulated, or has no analytical expression. Finite differences approximate derivatives using only these values, typically from a few nearby points. Consider the three points x_{j-1}, x_j, and x_{j+1} sketched in Fig. 2.1, with spacings $\Delta x_j = x_j - x_{j-1}$ and $\Delta x_{j+1} = x_{j+1} - x_j$. The grid need not be uniform, so generally $\Delta x_j \neq \Delta x_{j+1}$. Using the values $\phi_j = \phi(x_j)$, three common finite difference approximations of $d\phi/dx$ at x_j are:

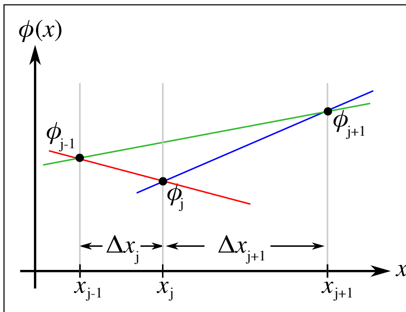

Fig 2.1

- **Forward difference:** uses ϕ_j and ϕ_{j+1}:

$$\frac{d\phi_j}{dx} \approx \frac{\phi_{j+1} - \phi_j}{\Delta x_{j+1}}, \tag{2.1}$$

  producing the slope in blue in Fig. 2.1.

- **Backward difference:** uses ϕ_j and ϕ_{j-1}:

$$\frac{d\phi_j}{dx} \approx \frac{\phi_j - \phi_{j-1}}{\Delta x_j}, \tag{2.2}$$

  producing the slope in red.

- **Centered difference:** uses ϕ_{j-1} and ϕ_{j+1}:

$$\frac{d\phi_j}{dx} \approx \frac{\phi_{j+1} - \phi_{j-1}}{\Delta x_j + \Delta x_{j+1}}, \tag{2.3}$$

  producing the slope in green.

  Note: Though called "centered," x_j is not necessarily midway between x_{j-1} and x_{j+1}.

When approximating derivatives in CFD, two key considerations are the *stencil* and *order-of-accuracy*. The *stencil* refers to the number and spatial distribution of the grid points used to approximate a derivative. The forward, backward, and centered differences all use two-point stencils, but differ in arrangement. The forward and backward differences bias information to one side of x_j, while centered differences use points on both sides. Later, we will see that advection terms often use stencils biased upstream, because advection propagates information downstream. By contrast, centered differences are typically used for diffusion, which propagates information in both directions.

The *order-of-accuracy* describes how the error in an approximation decreases as the grid is refined. For example, we can find the order-of-accuracy of the forward difference formula (2.1) by expanding ϕ_{j+1} about x_j in a Taylor series,

$$\phi_{j+1} = \phi_j + \Delta x_{j+1}\frac{d\phi_j}{dx} + \frac{\Delta x_{j+1}^2}{2}\frac{d^2\phi_j}{dx^2} + \frac{\Delta x_{j+1}^3}{6}\frac{d^3\phi_j}{dx^3} + ... \tag{2.4}$$

Solving for $d\phi_j/dx$ gives

$$\frac{d\phi_j}{dx} = \frac{\phi_{j+1} - \phi_j}{\Delta x_{j+1}} - \underbrace{\frac{\Delta x_{j+1}}{2}\frac{d^2\phi_j}{dx^2}}_{\text{dominant error term}} - \frac{\Delta x_{j+1}^2}{6}\frac{d^3\phi_j}{dx^3} - \ldots \tag{2.5}$$

As $\Delta x_{j+1} \to 0$, the dominant error varies linearly with Δx_{j+1}, often written as

$$\frac{d\phi_j}{dx} = \frac{\phi_{j+1} - \phi_j}{\Delta x_{j+1}} + \mathcal{O}(\Delta x_{j+1}), \tag{2.6}$$

where the "Big O" notation indicates the error is of order Δx_{j+1}. This leads to the following definitions of *truncation error* and *order-of-accuracy*.

Definition 2.1: Truncation Error

The *truncation error* of a finite-difference approximation is the difference between the exact derivative and the finite-difference estimate that arises from neglecting higher-order terms in the Taylor-series expansion. In Eq. (2.1), the dominant truncation error is $-\frac{\Delta x_{j+1}}{2}\frac{d^2\phi_j}{dx^2}$, which scales linearly with Δx_{j+1}.

Definition 2.2: Order-of-Accuracy

An approximation of order p has an error that varies like

$$Err \sim h^p, \tag{2.7}$$

where h is some measure of the grid spacing.

We can similarly determine the order-of-accuracy of the backward difference formula (2.2) by expanding ϕ_{j-1} about x_j in a Taylor series. This shows that the backward difference formula is also first-order-accurate,

$$\frac{d\phi_j}{dx} = \frac{\phi_j - \phi_{j-1}}{\Delta x_j} + \frac{\Delta x_j}{2}\frac{d^2\phi_j}{dx^2} + \ldots \tag{2.8}$$

To determine the order-of-accuracy of the centered difference formula (2.3), we expand both ϕ_{j+1} and ϕ_{j-1} about x_j, as below,

$$\begin{aligned} \phi_{j+1} &= \phi_j + \Delta x_{j+1}\tfrac{d\phi_j}{dx} + \tfrac{\Delta x_{j+1}^2}{2}\tfrac{d^2\phi_j}{dx^2} + \tfrac{\Delta x_{j+1}^3}{6}\tfrac{d^3\phi_j}{dx^3} + \ldots \\ \phi_{j-1} &= \phi_j - \Delta x_j\tfrac{d\phi_j}{dx} + \tfrac{\Delta x_j^2}{2}\tfrac{d^2\phi_j}{dx^2} - \tfrac{\Delta x_j^3}{6}\tfrac{d^3\phi_j}{dx^3} + \ldots \end{aligned} \tag{2.9}$$

Taking the difference of these equations, and solving for $d\phi_j/dx$, we find that

$$\frac{d\phi_j}{dx} = \frac{\phi_{j+1} - \phi_{j-1}}{x_{j+1} - x_{j-1}} + \underbrace{\frac{1}{2}(\Delta x_{j+1} - \Delta x_j)\frac{d^2\phi_j}{dx^2}}_{\text{dominant error term}} + \ldots \tag{2.10}$$

The dominant error term for the centered difference is proportional to the difference $\Delta x_{j+1} - \Delta x_j$, making it generally first-order accurate. However, for the special case of a uniform grid with $\Delta x_{j+1} = \Delta x_j = \Delta x$, the first-order error disappears,

$$\frac{d\phi_j}{dx} = \frac{\phi_{j+1} - \phi_{j-1}}{2\Delta x} + \underbrace{\frac{\Delta x^2}{6} \frac{d^3\phi_j}{dx^3}}_{\text{dominant error term}} + \ldots, \tag{2.11}$$

and the approximation is second-order accurate.

Even on non-uniform grids, the centered difference formula is typically more accurate than the forward and backward differences. To explain why, we can express the dominant error for the centered difference formula as

$$\frac{d\phi_j}{dx} = \frac{\phi_{j+1} - \phi_{j-1}}{x_{j+1} - x_{j-1}} + \underbrace{\frac{1}{2}(r-1)\,\Delta x_j \frac{d^2\phi_j}{dx^2}}_{\text{dominant error term}} + \ldots, \tag{2.12}$$

where $r = \Delta x_{j+1}/\Delta x_j$. We see that the dominant error term for the centered difference is equivalent to that of the backward difference in Eq. (2.8), but with the added factor $(r-1)$. If the grid spacing changes gradually, $(r-1)$ will be small, and the centered difference produces a much smaller error than the forward or backward difference. In fact, on slowly varying non-uniform grids, the centered difference formula typically produces nearly second-order accuracy.

Side Note 2.1: The practical use of finite differences in finite-volume methods

Finite-volume methods mainly use the forward, backward, and centered differences introduced above. This may surprise readers familiar with finite-difference PDE solvers, which approximate the differential form of conservation laws, such as that below for the advection–diffusion equation from Chapter 1,

$$\frac{\partial}{\partial t}(\rho\phi) + \nabla \cdot (\rho\phi\mathbf{u}) = \nabla \cdot (\Gamma\nabla\phi)\,. \tag{2.13}$$

These differential forms require estimates for first- and second-order derivatives. In contrast, finite-volume methods approximate the integral form of conservation laws,

$$\frac{d}{dt}\int_{CV} \rho\phi\, d\forall + \int_{CS} (\rho\phi\mathbf{u}) \cdot \mathbf{n}\, dA = \int_{CS} (\Gamma\nabla\phi) \cdot \mathbf{n}\, dA, \tag{2.14}$$

in which the highest derivative is first order.

2.2 Estimating error and order-of-accuracy

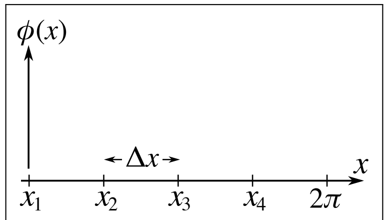

Fig 2.2

Two key questions in CFD are how to estimate the error of a numerical approximation, and how to estimate the order-of-accuracy of the method that produced the approximation. How you approach these questions is often problem dependent, and also depends on whether you are a developer with access to the underlying codes, or an end-user using CFD software as a black box.

For the scope of this initial discussion, suppose we are developing a code whose purpose is the approximation of the first derivative of a function $\phi(x)$. As input, the code takes values ϕ_j of the function at the following N points in the interval $0 \le x < 2\pi$,

$$x_j = (j-1)\Delta x, \qquad \Delta x = \frac{2\pi}{N}, \quad j = 1, 2, \dots, N. \tag{2.15}$$

The code then outputs an approximation of $d\phi/dx$ at each of the N points. These are computed using both the forward difference formula (2.1) and the centered difference formula (2.3). Figure 2.2 shows the grid points when $N = 4$. The code assumes the function is periodic over 2π, and does not include a grid point at $x = 2\pi$ because periodicity gives $\phi(2\pi) = \phi(0)$.

As a developer, we can test the code against a function with a known exact derivative. For that, we choose the function

$$\phi(x) = \exp(\sin x). \tag{2.16}$$

Using this test function, File (2.1) approximates $d\phi/dx$ at each x_j using forward and centered differences. The results are stored in the arrays `FD` and `CD`.

File 2.1: A Matlab function that computes forward and centered finite-difference approximations and their errors

```
function [FDerr,CDerr] = FiniteDiff(N)
% Finite differences of phi(x) = exp(sin(x)) on a uniform grid

% Grid and function
dx = 2*pi/N;                   % grid spacing
x  = (0:N-1)'*dx;              % grid points
F  = exp(sin(x));              % function phi(x)
FE = cos(x).*exp(sin(x));    % exact derivative

% Forward difference
FD = zeros(N,1);
for i = 1:N-1
    FD(i) = (F(i+1) - F(i))/dx;
end
FD(N) = (F(1) - F(N))/dx;     % apply periodic BC

% Centered difference
CD = zeros(N,1);
for i = 2:N-1
    CD(i) = (F(i+1) - F(i-1))/(2*dx);
end
CD(1) = (F(2) - F(N))/(2*dx);
CD(N) = (F(1) - F(N-1))/(2*dx);
```

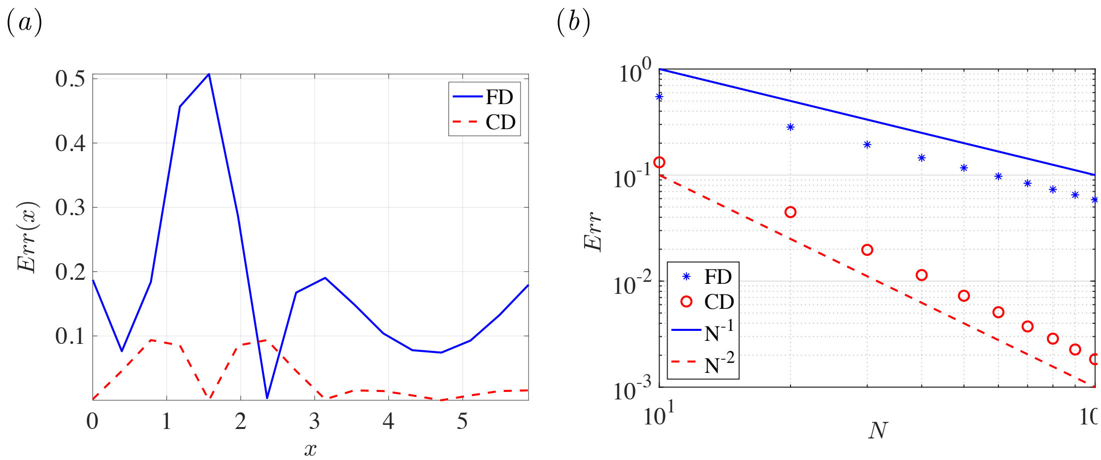

Fig 2.3: (a) The variation of the local error e_j with x_j produced by the forward (blue) and centered (red) difference formulas when $N = 16$. (b) A log–log plot of the global error E_∞ versus N generated by the forward (blue) and centered (red) difference formulas.

```
24
25 % compute local absolute error
26 FDerr = abs(FE-FD);
27 CDerr = abs(FE-CD);
28
29 % compute global relative error using infinity norm
30 FDERR = max(abs(FE-FD))/max(abs(FE));
31 CDERR = max(abs(FE-CD))/max(abs(FE));
```

Even with an exact solution, an important question arises: how should errors be quantified? One way is to examine the local (pointwise) error at each grid point,

$$e_j = \left|g_e(x_j) - g_j\right|, \tag{2.17}$$

where $g_e(x_j)$ is the exact value and g_j is its numerical approximation. The local error e_j is computed on lines 26–27 of File (2.1). This produces plots like Fig. 2.3(a), showing e_j versus x_j for the forward (blue) and centered (red dashed) differences when $N = 16$.

In addition to the local error e_j, we often need some overall measure of the "global" error (E). In this textbook, we use the measure below,

$$E_\infty = \frac{\|\mathbf{g}_e - \mathbf{g}_N\|_\infty}{\|\mathbf{g}_e\|_\infty}, \tag{2.18}$$

where $\mathbf{g}_e$ and $\mathbf{g}_N$ are vectors containing the exact and numerical values of the quantity of interest at all grid points x_j. The symbol $\|\cdot\|_\infty$ denotes the *infinity norm*, which is equivalent to the maximum absolute value of a vector. This calculation is implemented on lines 30–31 of File (2.1). Because the error is normalized by the infinity norm of the exact solution, a value such as $E_\infty = 0.005$ corresponds to a relative error of 0.5%.

The order-of-accuracy of a numerical method is typically assessed by plotting the global error E versus N, using logarithmic scales on both axes. To explain why, suppose that a method is p-order accurate on a uniform grid. In that case, its global error should scale like

$$E \sim \Delta x^p. \tag{2.19}$$

If the computational domain has width L, then the grid spacing is $\Delta x = L/N$, which allows us to write

$$E \sim N^{-p}. \tag{2.20}$$

Taking the logarithm of both sides gives

$$\log(E) \sim -p \log(N). \tag{2.21}$$

As a result, a log–log plot of E versus N should produce a straight line with slope $-p$.

Figure 2.3(b) shows such a plot for the forward and centered difference approximations produced by the Matlab function `FiniteDiff.m`. The error for the forward difference decreases like $1/N$ (shown as a solid line), while that for the centered difference decreases like $1/N^2$ (shown as a dashed line). This confirms the expected orders-of-accuracy for both methods.

Side Note 2.2: Other common measures of global error

In addition to the infinity norm used in this course, global errors are often reported using the *1-norm* and *2-norm*. Unfortunately, these terms are used inconsistently in the literature, because their definitions depend on whether you interpret error as a function defined over a physical domain, or as a vector arising from a discrete numerical method.

First, consider the notion of a norm applied to a function. Suppose we want to measure how "large" some function $\phi(x)$ is on the interval $x \in [a, b]$. Two common measures are

$$\|\phi(x)\|_1 = \int_a^b |\phi(x)|\, dx, \qquad \|\phi(x)\|_2 = \left[\int_a^b |\phi(x)|^2\, dx\right]^{1/2}. \tag{2.22}$$

In our case, because we only know the error at discrete grid points x_j, we must approximate the above integrals numerically. Using the left-hand rule produces,

$$E_1 = \sum_{j=1}^{N} \Delta x_j e_j, \qquad E_2 = \left[\sum_{j=1}^{N} \Delta x_j e_j^2\right]^{1/2}. \tag{2.23}$$

On a uniform grid with $\Delta x = L/N$, these simplify to

$$\boxed{E_1 = \frac{L}{N}\sum_{j=1}^{N} e_j, \qquad E_2 = \left[\frac{L}{N}\sum_{j=1}^{N} e_j^2\right]^{1/2}.} \tag{2.24}$$

An alternate set of definitions arises in the linear algebra community, where the quantity of interest is a vector rather than a function. In that case, the 1-norm and 2-norm of a vector $\mathbf{u}$ are defined as

$$\|\mathbf{u}\|_1 = \sum_{j=1}^{N} |u_j|, \qquad \|\mathbf{u}\|_2 = \left[\sum_{j=1}^{N} |u_j|^2\right]^{1/2}. \tag{2.25}$$

Here, the factor L/N is omitted, and the 2-norm corresponds to the Euclidean length of the vector. When reading CFD literature, it is common to encounter error measures based on these vector norms, often without explicit definitions. This ambiguity can make results difficult to interpret. In this textbook, we prefer the infinity norm because it has a single, unambiguous definition and represents the worst error encountered anywhere in the domain.

2.3 Optional reading: additional finite difference formulas

Here we present some additional finite-difference formulas that can be useful.

2.3.1 A second-order accurate approximation on non-uniform grids

Using Taylor series, we can derive a finite difference approximation to $d\phi/dx$ that uses information at three points to formally maintain second-order accuracy on non-uniform grids. For that, we expand both ϕ_{j+1} and ϕ_{j-1} about x_j,

$$
\begin{aligned}
\phi_{j+1} &= \phi_j + \Delta x_{j+1}\frac{d\phi_j}{dx} + \frac{\Delta x_{j+1}^2}{2}\frac{d^2\phi_j}{dx^2} + \frac{\Delta x_{j+1}^3}{6}\frac{d^3\phi_j}{dx^3} + \ldots \\
\phi_{j-1} &= \phi_j - \Delta x_j\frac{d\phi_j}{dx} + \frac{\Delta x_j^2}{2}\frac{d^2\phi_j}{dx^2} - \frac{\Delta x_j^3}{6}\frac{d^3\phi_j}{dx^3} + \ldots
\end{aligned}
\tag{2.26}
$$

and we then seek a sum of these two equations that cancels the terms proportional to Δx^2, which are highlighted in red above. The required sum is

$$
\Delta x_j^2\phi_{j+1} - \Delta x_{j+1}^2\phi_{j-1} = \left(\Delta x_j^2 - \Delta x_{j+1}^2\right)\phi_j + \left(\Delta x_j^2\Delta x_{j+1} - \Delta x_{j+1}^2\Delta x_j\right)\frac{d\phi_j}{dx} + \ldots
\tag{2.27}
$$

After some reorganization, this produces

$$
\frac{d\phi_j}{dx} = \frac{\Delta x_j^2\phi_{j+1} - \Delta x_{j+1}^2\phi_{j-1} - \left(\Delta x_j^2 - \Delta x_{j+1}^2\right)\phi_j}{\Delta x_j\Delta x_{j+1}\left(\Delta x_j + \Delta x_{j+1}\right)} - \underbrace{\frac{\Delta x_j\Delta x_{j+1}}{6}\frac{d^3\phi_j}{dx^3}}_{\text{second-order}}.
\tag{2.28}
$$

This formula reduces to the centered-difference formula on a uniform grid. Though this formula is second-order accurate on any grid, we will see later that it is not typically used in finite-volume methods.

2.3.2 One-sided differences

Suppose you only know the value of $\phi(x)$ at the four equispaced locations marked in Fig. 2.4. For the interior points x_2 and x_3, you can estimate $d\phi/dx$ to second-order accuracy using the centered-difference formula. At the points x_1 and x_4, however, the centered-difference formula is not applicable, because we only have data to one side of the point. To maintain second-order accuracy, we must derive a second-order one-sided difference. For example, to estimate $d\phi/dx$ at x_1, we expand ϕ_2 and ϕ_3 in a Taylor series about x_1,

$$
\begin{aligned}
\phi_2 &= \phi_1 + \Delta x\frac{d\phi_1}{dx} + \frac{\Delta x^2}{2}\frac{d^2\phi_1}{dx^2} + \mathcal{O}(\Delta x^3) \\
\phi_3 &= \phi_1 + 2\Delta x\frac{d\phi_1}{dx} + 2\Delta x^2\frac{d^2\phi_1}{dx^2} + \mathcal{O}(\Delta x^3)
\end{aligned}
$$

We then see that we can cancel the $d^2\phi_1/dx^2$ terms by taking the linear combination $4\phi_2 - \phi_3$,

$$4\phi_2 - \phi_3 = 3\phi_1 + 2\Delta x \frac{d\phi_1}{dx} + \mathcal{O}(\Delta x^3), \tag{2.29}$$

which can be solved for $d\phi_1/dx$,

$$\frac{d\phi_1}{dx} = \frac{-3\phi_1 + 4\phi_2 - \phi_3}{2\Delta x} + \mathcal{O}(\Delta x^2). \tag{2.30}$$

Following a similar approach, it can be shown that $d\phi_4/dx$ can be estimated to second order as

$$\frac{d\phi_4}{dx} = \frac{3\phi_4 - 4\phi_3 + \phi_2}{2\Delta x} + \mathcal{O}(\Delta x^2). \tag{2.31}$$

In the interest of generality, these relations can be written as

$$\boxed{\frac{d\phi_j}{dx} \approx \frac{-3\phi_j + 4\phi_{j+1} - \phi_{j+2}}{2\Delta x}, \qquad \frac{d\phi_j}{dx} \approx \frac{3\phi_j - 4\phi_{j-1} + \phi_{j-2}}{2\Delta x}.} \tag{2.32}$$

2.3.3 Approximating the second derivative

While a minimum of two points is required to approximate a first derivative, a minimum of three is required for the second derivative. Suppose we know the function $\phi(x)$ at the locations x_{j-1}, x_j, and x_{j+1} in Fig. 2.1. To approximate $d^2\phi/dx^2$ at x_j, we expand ϕ_{j-1} and ϕ_{j+1} in Taylor series about x_j,

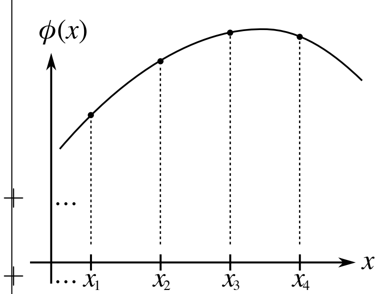

Fig 2.4

$$\begin{aligned}
\phi_{j+1} &= \phi_j + \Delta x_{j+1}\frac{d\phi_j}{dx} + \frac{\Delta x_{j+1}^2}{2}\frac{d^2\phi_j}{dx^2} + \frac{\Delta x_{j+1}^3}{6}\frac{d^3\phi_j}{dx^3} + \dots \\
\phi_{j-1} &= \phi_j - \Delta x_j\frac{d\phi_j}{dx} + \frac{\Delta x_j^2}{2}\frac{d^2\phi_j}{dx^2} - \frac{\Delta x_j^3}{6}\frac{d^3\phi_j}{dx^3} + \dots
\end{aligned}$$

We then seek a sum of these two equations that cancels the terms proportional to $d\phi_j/dx$, which are highlighted in red above. The required sum is

$$\Delta x_j\phi_{j+1} + \Delta x_{j+1}\phi_{j-1} = (\Delta x_j + \Delta x_{j+1})\,\phi_j + \frac{1}{2}\left(\Delta x_j \Delta x_{j+1}^2 + \Delta x_{j+1}\Delta x_j^2\right)\frac{d^2\phi_j}{dx^2} + \dots$$

After some reorganization, this produces the following first-order accurate estimate of $d^2\phi/dx^2$,

$$\frac{d^2\phi_j}{dx^2} = \frac{\Delta x_{j+1}\phi_{j-1} - (\Delta x_j + \Delta x_{j+1})\,\phi_j + \Delta x_j\phi_{j+1}}{\frac{1}{2}\Delta x_j\Delta x_{j+1}\,(\Delta x_j + \Delta x_{j+1})} + \underbrace{\frac{1}{3}(\Delta x_{j+1} - \Delta x_j)\frac{d^3\phi_j}{dx^3}}_{\text{first-order}}. \tag{2.33}$$

On a uniform grid, this formula simplifies to the following second-order accurate result,

$$\frac{d^2\phi_j}{dx^2} = \frac{\phi_{j-1} - 2\phi_j + \phi_{j+1}}{\Delta x^2} + \mathcal{O}(\Delta x^2). \tag{2.34}$$

2.4 Summary of key concepts

Key definitions:

1. **Finite-difference approximation:** a numerical estimate of a derivative using values at discrete grid points.
2. **Stencil:** the set of points used to compute a numerical approximation, including number and arrangement. Stencils also apply to numerical interpolation and integration schemes.
3. **Forward, backward, and centered differences:** common two-point formulas for first-order derivatives, using one-sided (forward/backward) or symmetric (centered) information.
4. **Truncation error:** error arising from neglecting higher-order terms in a Taylor-series expansion.
5. **Order-of-accuracy:** a measure of how rapidly the error of a numerical approximation decreases as the grid is refined.

Key concepts:

1. Stencil arrangement affects accuracy and directional bias.
2. On non-uniform grids, centered differences are formally first-order but often more accurate than one-sided differences.
3. Numerical error can be measured locally or globally. When reporting global errors, it is important to clearly define whatever norm is used. We recommend using the infinity norm.
4. Observed order-of-accuracy is commonly estimated using log–log plots of global error versus grid resolution.